UPV (Ultrasonic Pulse Velocity) — Test, Interpretation, and OUANTUM Automation Requirements


Sprint 1 · Weeks 1–2
Data foundation + input layer
Project + reading intake
Name, grade, date cast, age
Reading entry
Velocity, distance, travel time
Postgres schema
projects · test_points · readings · devices · operators
✓ Output: structured data stored, API endpoints live
Sprint 2 · Weeks 3–4
Validation engine (Layer 2)
Auto-classify
Excellent / Good / Medium / Doubtful
Formula check
V = L / T cross-check
flag outliers > 2σ
Score output
Quality grade
Confidence + Risk 0–100
✓ Output: every reading gets a grade, outliers flagged before analysis
Sprint 3 · Weeks 5–7
AI reasoning — defect detection (Layer 3)
Claude API call — defect classifier
Input: readings + stats · Output: defect type, confidence, reasoning
Rule-based triggers
Sudden drop → honeycombing flag
Plain-language reasoning
"P14 shows 22% drop — possible void"
✓ Output: AI narrative per test point, defect probability 0–100%
Sprint 4 · Weeks 8–9
Monitoring + spatial engine (Layers 4 & 5)
Time-series store
Day 7 / 14 / 28 readings
Growth rate, stability index
Heatmap grid
Map each point to element
Green / Yellow / Red zones
Dashboard KPIs
Avg / min / max velocity · uniformity index · quality score · failed locations
✓ Output: live dashboard, trend charts, spatial heatmap view
Sprint 5 · Weeks 10–11
Report generation engine (Layer 6)
Auto-report builder
QA/QC + inspection + summary
Engineer approval flow
Review → annotate → sign off
✓ Output: downloadable report, audit trail, engineer sign-off
Phase 1 complete → scale to Phase 2

QA/QC engine

Velocity classification

$V = L / T$ · km/s

>4.5 Excellent · 3.5–4.5 Good

3.0–3.5 Medium · <3.0 Doubtful

Uniformity index

$UI = (\sigma / \bar{V}) \times 100$

σ = std dev · $\bar{V}$ = mean velocity

UI <5% Uniform · >15% Non-uniform

Quality score (0–100)

$QS = (V\_norm \times 50) + (UI\_norm \times 30)$

$+ (age\_factor \times 20)$

Normalize each factor to 0–1

Outlier detection

$Z = (x - \bar{V}) / \sigma$

$|Z| > 2$ → flag as outlier

IQR method as fallback

Structural integrity score

$SIS = QS \times 0.5 + DefectPenalty \times 0.3$

$+ TrendBonus \times 0.2$

DefectPenalty = sum of defect weights

Reliability score

$RS = (n / n\_req) \times 40$

$+ calibration\_ok \times 40 + signal\_ok \times 20$

n_req = min readings per element

Coefficient of variation

$CoV = (\sigma / \bar{V}) \times 100$

<10% Good · 10–20% Fair · >20% Poor

Range ratio

$RR = (V\_max - V\_min) / \bar{V}$

RR >0.3 → high internal variation

Confidence score

Based on: RS + consistency + grade

Weighted average → 0–100%

Risk score

$Risk = 100 - QS$

Adjust by element criticality weight

Failed locations count

V <3.0 km/s → Failed

V <2.5 km/s → Critical

Pass / fail ratio

$PFR = passed\_count / total\_count$

PFR <0.7 → escalate to engineer

Age correction factor

Expected V at age t: use strength-gain curve (IS 456 / ACI 209) · compare actual vs expected → deviation %

Workmanship defect engine

Honeycombing
Local drop: (V_point − V_neighbours) / V_neighbours
>15% drop → suspect · >25% → high confidence
Neighbours = adjacent 4 test points

Voids
Major velocity reduction: V <2.5 km/s
Sudden single-point low in a zone
Confirmed if Z-score >3 for that point

Segregation
Irregular spatial pattern across element
CoV >20% + no clear directional trend
High variance without single-point outlier

Poor compaction
Consistently low readings across element
V̄ <3.5 · CoV <15% (uniform but low)
No outlier — whole element is weak

Cold joints
Sharp velocity step across layer boundary
ΔV >0.5 km/s between adjacent layers
Map test points to pour sequence zones

Cracks
Path discontinuity: travel time spikes
T_actual >> T_expected for given L
Indirect method shows higher sensitivity

Excess water
Early-age V lower than grade benchmark
Day-7 V <expected by >10%
Slow gain rate over 7→14 days

Inadequate curing
Surface points lower than core points
ΔV(surface−core) >0.3 km/s
Age-mismatch: gain stalls after day 7

Defect probability score (per defect)
P(defect) = rule_trigger_weight × 0.5 + Z_score_norm × 0.3 + AI_confidence × 0.2
Output: 0–100% for each defect type · highest score = primary diagnosis

Composite defect index
CDI = weighted sum of all active defect probabilities · cap at 100 · feeds into SIS penalty

Overview

UPV (Ultrasonic Pulse Velocity) is a **Non-Destructive Testing (NDT)** method used to evaluate the **quality, uniformity, and integrity of concrete** by measuring the travel speed of an ultrasonic pulse through the material.

What UPV measures

- **Transit time (T):** the time taken by the pulse to travel between probes
- **Path length (L):** the distance between probes along the travel path
- **Pulse velocity (V):** an indicator that correlates with density, elastic properties, and presence of discontinuities

Principle and Formula

An ultrasonic pulse is generated by a **transmitting transducer (Tx)** and received by a **receiving transducer (Rx)**.

$$V = \frac{L}{T}$$

Where:

- **V** = pulse velocity (m/s or km/s)
- **L** = path length (m)
- **T** = transit time (s)

Example Calculation

Given:

- **L = 0.5 m**
- **T = 125 μs = 125 × 10^{-6} s**

$$V = \frac{0.5}{125 \times 10^{-6}} = 4000 \text{ m/s} = 4.0 \text{ km/s}$$

Interpretation (typical): **Good quality concrete**.

Test Arrangements (Probe Placement)

1) Direct Transmission (best accuracy)

```
Tx  ------------ Concrete ------------ Rx
```

- Highest accuracy
- Strongest signal
- Preferred when both faces are accessible

2) Semi-direct Transmission

```
 Tx
|
Concrete
|
Rx
```

- Used when opposite faces are not accessible
- Moderate accuracy

3) Indirect (surface) Transmission

`Tx ------- Surface ------- Rx`

- Used when only one face is accessible
- Least accurate
- Strongly influenced by surface condition

Quality Classification (Typical)

Velocity (km/s)	Concrete quality (typical)
> 4.5	Excellent
3.5 – 4.5	Good
3.0 – 3.5	Medium
< 3.0	Doubtful or Poor

Velocity limits vary by standard, aggregate type, moisture condition, member geometry, and project acceptance criteria.

Factors Affecting UPV (What changes V)

Increases velocity (generally)

- Higher density and better compaction
- Higher strength and better quality paste-aggregate bonding
- Proper curing
- Lower crack density

Decreases velocity (generally)

- Cracks, voids, honeycombing
- Segregation and poor compaction
- Deterioration (freeze-thaw, chemical attack)
- Poor curing

⚠️ Moisture can significantly change readings: wet concrete often shows higher UPV than dry concrete. Always record moisture condition and compare like-for-like.

How UPV is Used in QA, QC, and Monitoring

QA (Quality Assurance)

- Confirms process consistency across pours
- Checks workmanship quality and curing effectiveness
- Supports compliance evidence for specifications

QC (Quality Control)

- Detects internal defects (voids, honeycombing)
- Compares quality across elements and repairs
- Flags suspicious zones for further tests

Structural monitoring

- Baseline assessment (initial health)
- Periodic monitoring (trend of deterioration)
- Post-event assessment (fire, impact, abnormal loading)

Advantages and Limitations

Advantages

- Non-destructive and fast
- Portable equipment
- Helps identify hidden defects
- Cost-effective compared with extensive coring

Limitations

- Does not directly measure compressive strength

- Requires surface preparation and consistent coupling
- Affected by moisture, temperature, and reinforcement
- Best used with other tests and engineering judgment

Standards (Common)

- **ASTM C597**
- **IS 13311 (Part 1)**
- **BS 1881**

Typical Field Workflow (Good Practice)

1. Identify element and test grid (location map)
2. Verify device calibration and couplant
3. Measure and record **L** accurately
4. Measure and record **T** (multiple readings recommended)
5. Compute **V = L/T** automatically
6. Compare with acceptance criteria (project-specific)
7. If abnormal, confirm using complementary tests
    - Rebound hammer
    - Core test
    - Visual inspection
    - GPR or other NDT where applicable
8. Generate report with traceability and approvals

OUANTUM: Turning UPV into an AI-driven QA/QC Decision System

What happens today (pain points)

- Manual readings and data entry

- Human error and inconsistent formats
- No centralized history
- Difficult trend analysis
- Weak decision support (engineers interpret manually)

OUANTUM Vision: 3-phase capability

Phase 1 — Digital Calculation Engine (must-have MVP)

Inputs

- Structure ID and element type
- Location and test point ID
- Method (direct, semi-direct, indirect)
- Path length (L)
- Transit time (T)

System outputs

- Auto-calculated UPV
- Auto-classified quality band (project-specific thresholds)
- Standardized QA/QC report
- Cloud storage and searchable history

Phase 2 — AI interpretation (decision support)

Beyond velocity, provide:

- Change detection vs baseline or previous readings
- Probable causes (cracking, voids, moisture variation, coupling issues)
- Recommended next actions (retest, add points, confirm with other NDT)
- Confidence score and explanation (what signals drove the recommendation)

Phase 3 — Defect prediction and risk scoring

Combine UPV with:

- Rebound hammer readings
- Core results (where available)
- Concrete age and curing history
- Mix design, cement type, aggregate type
- Environmental and exposure conditions
- Element geometry and reinforcement density

Outputs:

- Probability of honeycombing or voids
- Probability of strength deficiency (with uncertainty)
- Structural risk score
- Maintenance and inspection recommendations

Digital Twin and Visualization (optional but high value)

- Map every test point to an element coordinate system
- Color-coded health map by threshold
- Trend charts per element, pour, floor, or zone
- Alerts when degradation rate exceeds limits

Requirements and Data Needed (Add this to make the system real)

A) Field data capture requirements

Required (minimum)

- Project, structure, element ID
- Test point ID and coordinates (or grid reference)
- Date and time
- Operator and organization

- Transmission method (direct, semi-direct, indirect)
- Path length **L** and unit
- Transit time **T** and unit
- Number of repetitions and averaged value
- Device model and calibration status

Recommended (for accuracy and AI)

- Surface condition and preparation notes
- Couplant type
- Member thickness and geometry
- Estimated reinforcement presence along path (yes or no, or bar map if available)
- Concrete age at test time (days)
- Moisture condition (wet, dry, saturated-surface-dry)
- Ambient and surface temperature
- Site photos of test setup (Tx and Rx placement)

B) Reference and acceptance criteria requirements

- Project-specific threshold table (often differs from generic bands)
- Baseline readings per pour or element
- Retest rules
    - When variance exceeds X%
    - Minimum points per element
    - Required replication count
- Escalation workflow rules (when to request rebound hammer or core)

C) Supporting datasets (for Phase 2 and Phase 3)

Concrete and construction metadata

- Mix design details (w/c ratio, cement type, admixtures)
- Pour ID and pour date
- Curing method and duration
- Cube or cylinder test results (strength vs age)
- Formwork removal date and loading history (if available)

Environment and exposure

- Weather history during curing (temperature, rainfall, humidity)
- Exposure class (marine, industrial, freeze-thaw, etc.)

Complementary testing data

- Rebound hammer readings (with mapping to same points)
- Core test results (compressive strength, density)
- Visual defect logs and repair history

D) System requirements (product and engineering)

Workflow and usability

- Offline-first mobile capture (site connectivity is inconsistent)
- Standard forms with validation (units, range checks, mandatory fields)
- Auto-calculation and auto-classification
- Role-based approvals (engineer review and sign-off)
- Versioned reports (audit trail)

Data architecture and governance

- Centralized database schema with strict identifiers (project → structure → element → point)
- File storage for photos and attachments
- Audit logs for edits and approvals
- Data retention policy and export support

AI requirements (practical)

- Labelled historical dataset (defect confirmed or not, by core or inspection)
- Ground-truth linking (UPV points ↔ confirmed defect locations)
- Model monitoring (drift checks across regions, aggregates, device types)
- Explainability (why a zone was flagged)

Integrations (optional but important)

- Import from drawings or BIM (point mapping)
- Integration with QA/QC checklists and NCR workflow
- APIs to pull lab results (cube tests) and push reports to clients

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E) What is commonly missed (add-ons that reduce failure)

- **Calibration workflow**: reminders, calibration certificates upload, lockout if expired
- **Re-test logic**: auto-flag outliers and request replicate readings
- **Uncertainty handling**: report variance, not just a single value
- **Point mapping discipline**: without consistent point IDs, trend monitoring fails
- **Training content**: built-in SOP, videos, and checklists for correct probe placement

> ✅ If you want, I can convert the requirements section into a clean checklist that can be used as an OUANTUM product spec for UPV automation.